

# Omkar Swami

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GitHub | LinkedIn | LeetCode | Codeforces | TryHackMe

## INTRODUCTION

Backend and AI systems engineer building multimodal search pipelines, distributed automation tools, and secure APIs. Experienced with LLM orchestration, vector search, and Airflow-based data workflows.

## EDUCATION

**Parul University** Grad. 2026  
B.Tech in Computer Science and Engineering | CGPA: 7.93  
**Narayana College, Hyderabad** Grad. 2022  
Intermediate (PCM) | 83.7%  
**Times Public School, Udgir** Grad. 2020  
SSC | 90.8%

## INTERNSHIP EXPERIENCE

**Data Analyst Intern — Acadify Solutions** Feb 2026 – Present

- Building a Jira-inspired project management platform with Kanban boards, task lifecycle tracking, and role-based access control using PHP and MySQL.
- Designed normalized relational schemas for task assignment and multi-user workflows; consolidated legacy tables to reduce query complexity.

## PROJECTS

**Gnosis** — *Multimodal Semantic Search Engine* Python, FastAPI, Airflow, PostgreSQL, Qdrant, Docker, Ollama, SentenceTransformers

- Architected an end-to-end pipeline for natural language retrieval over unstructured documents and images (PDF, DOCX, PPTX, JPG/PNG) with MIME-validated file ingestion.
- Built fault-tolerant Airflow DAGs orchestrating chunking, BGE embedding, and vector indexing into Qdrant; metadata and file status tracked via PostgreSQL.
- Deployed full stack via Docker Compose: FastAPI (REST), Qdrant, PostgreSQL, Redis+Celery task queue, and GPU-enabled Airflow workers.

**Nous** — *AI-Assisted Distributed Systems Administration Tool* Python, C#, Avalonia UI, LLMs, RAG, Vector DB, TLS/RSA, Sockets

- Built a cross-platform remote administration tool with a desktop GUI (Avalonia) enabling natural language command execution across networked client listeners.
- Implemented TLS-encrypted socket channels with RSA key exchange; integrated RAG pipelines over a local vector DB for contextual, system-aware command suggestions.

## ACHIEVEMENTS & CERTIFICATIONS

- **TryHackMe Top 1%** globally (3M+ users) | **5th Place** — International FOSS CTF | **200+ DSA** problems solved.
- NPTEL: Computer Networks, Theory of Computation | Database & SQL — Infosys | Roadmap to Ethical Hacking — Parul University.
- Participant — Red Team Security Summit 2023, Kochi.

## SKILLS

**Languages:** Python, Java, C++, C#, PHP

**AI & Data:** LLM Orchestration, RAG Pipelines, Vector Search (Qdrant), SentenceTransformers, PySpark, NumPy, Pandas, Generative AI

**Backend & Infra:** FastAPI, Apache Airflow, Docker, PostgreSQL, MySQL, Redis, Celery, pytest

**Web:** MERN Stack, .NET / Avalonia

**Security & Networking:** TLS/RSA, Secure Socket Programming, Cybersecurity, Linux